CANONICAL MORPHISM FROM AN OBJECT INTO THE INTERNAL HOM OF THE DERIVED TENSOR PRODUCT WITH ITSELF

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete construction: the representatives, the quasi-isomorphism into a K-injective complex and the composite of the two displayed maps are all inside the counted span.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology and in chapter more-algebra. Those statements are the special cases of this ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The single labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- K-flat and K-injective complexes of O-modules. Same chapter, the definitions of K-flatness and K-injectivity and the existence of K-injective replacements. Supplies the representatives K^bullet representing K, L^bullet representing L and the K-injective J^bullet of the counted proof.
- SheafHom complex and RSheafHom on the derived category. Same chapter, the complex of sheaves of homomorphisms and the derived internal hom it computes for a K-injective target. Supplies the middle and right hand terms of the displayed chain and the identification of the target with RSheafHom(L, K tensor^L L).
- Derived tensor product computed by a K-flat representative. Same chapter, the derived tensor product and the role of K-flatness in computing it. Supplies that Tot(K^bullet tensor L^bullet) represents K tensor^L L, so that the internal hom into J^bullet computes the target.

Source reference. Original lemma 0A93: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8681–8681. Same-chapter navigation pointer invoked at proof line 9110, where the author takes the first of the two displayed maps from that earlier lemma on the diagonal map. The pointer only resolves the reference for the reader; the cited lemma is not reproduced inside the counted span, and no bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Given K, L in $D(\mathcal{O})$ there is a canonical morphism*

$$K \longrightarrow R\mathcal{H}om(L, K \otimes^{\mathbf{L}}_{\mathcal{O}} L)$$

in $D(\mathcal{O})$ functorial in both K and L.

Proof. Choose a K-flat complex $\mathcal{K}^\bullet$ representing K and any complex of $\mathcal{O}$-modules $\mathcal{L}^\bullet$ representing L. Choose a K-injective complex $\mathcal{J}^\bullet$ and a quasi-isomorphism $\mathrm{Tot}(\mathcal{K}^\bullet \otimes_{\mathcal{O}} \mathcal{L}^\bullet) \to \mathcal{J}^\bullet$. Then we use

$$\mathcal{K}^\bullet \to \mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathrm{Tot}(\mathcal{K}^\bullet \otimes_{\mathcal{O}} \mathcal{L}^\bullet)) \to \mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathcal{J}^\bullet)$$

where the first map comes from Lemma 0A93. □